



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 01

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5 , CSS3 , JavaScript , Visual Studio Code , Google Chrome

Experiment Program Code:

1. External CSS:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Experiment 1</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <div>
```

```
    <h1>Welcome to the Lab!</h1>
```

```
    <p><strong>Name:</strong> Aishwarya Brahmkar</p>
```

```
    <p><strong>PRN:</strong> 24070521212</p>
```

```
    <button onclick="alert('External CSS active!')">Test External CSS</button>
```

```
  </div>
```



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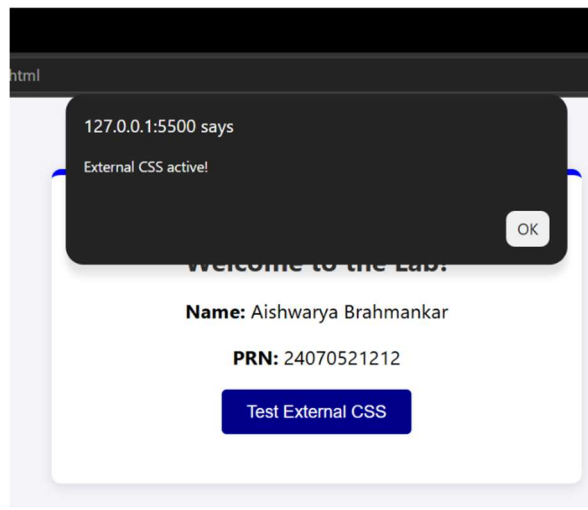
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</body>

</html>

Output:



2. Inline CSS:

<!DOCTYPE html>

<html>

<head>

<title>Experiment 1</title>

</head>

<body style="font-family: 'Segoe UI', sans-serif; background-color: #f4f4f9; display: flex; justify-content: center; padding-top: 50px; margin: 0;">

<div style="background-color: white; padding: 40px; border-radius: 8px; border-top: 5px solid blue; box-shadow: 0 2px 10px rgba(0,0,0,0.1); text-align: center; width: 350px;">



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<h1 style="color: #333; font-size: 22px;">Welcome to the Lab!</h1>

<p>Name: Aishwarya Brahmkar</p>

<p>PRN: 24070521212</p>

<button onclick="alert('Inline CSS active!')"

style="padding: 10px 20px; background-color: blue; color: white; border: none;
border-radius: 4px; cursor: pointer; font-size: 14px;">

Test Inline CSS

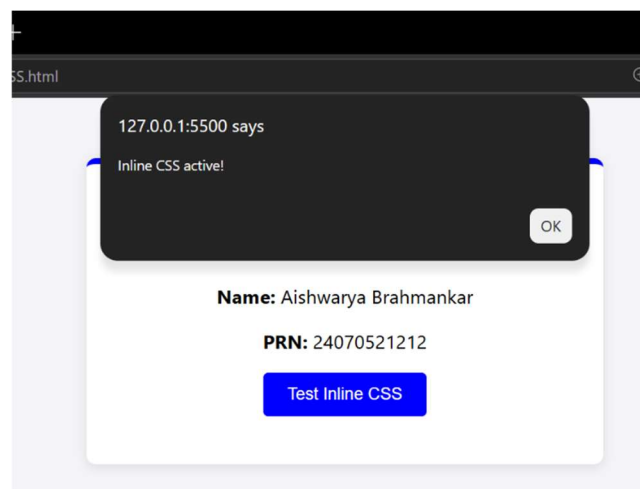
</button>

</div>

</body>

</html>

Output:





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3. Internal CSS:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Experiment 1</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: 'Segoe UI', sans-serif;
```

```
      background-color: #f4f4f9;
```

```
      display: flex;
```

```
      justify-content: center;
```

```
      padding-top: 50px;
```

```
      margin: 0;
```

```
    }
```

```
    div {
```

```
      background-color: white;
```

```
      padding: 40px;
```

```
      border-radius: 8px;
```

```
      border-top: 5px solid blue;
```

```
      box-shadow: 0 2px 10px rgba(0,0,0,0.1);
```



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```
text-align: center;

width: 350px;

}

h1 {

  color: #333;

  font-size: 22px;

}

button {

  padding: 10px 20px;

  background-color: blue;

  color: white;

  border: none;

  border-radius: 4px;

  cursor: pointer;

  font-size: 14px;

}

button:hover {

  background-color: darkblue;

}
```



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```
</style>

</head>

<body>

  <div>

    <h1>Welcome to the Lab!</h1>

    <p><strong>Name:</strong> Aishwarya Brahmkar</p>

    <p><strong>PRN:</strong> 24070521212</p>

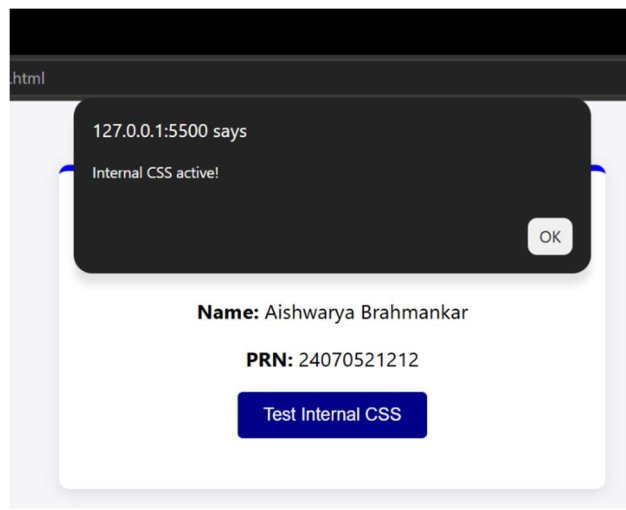
    <button onclick="alert('Internal CSS active!')">Test Internal CSS</button>

  </div>

</body>

</html>
```

Output:





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Style:

/* External CSS used to style the page */

body {

font-family: 'Segoe UI', sans-serif;

background-color: #f4f4f9;

display: flex;

justify-content: center;

padding-top: 50px;

}

div {

background-color: white;

padding: 40px;

border-radius: 8px;

border-top: 5px solid blue; /* Adds a nice accent */

box-shadow: 0 2px 10px rgba(0,0,0,0.1);

text-align: center;

width: 350px;

}

h1 {



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```
color: #333;

font-size: 22px;
}

button {
    padding: 10px 20px;
    background-color: blue;
    color: white;
    border: none;
    border-radius: 4px;
    cursor: pointer;
    font-size: 14px;
}

button:hover {
    background-color: darkblue;
}
```



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Case Study No.: 01

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5 , CSS3 , JavaScript , Visual Studio Code , Google Chrome

Experiment Program Code:

1. Main:

```
<html>
```

```
<head>
```

```
<title>SIT Nagpur - Department Page</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    margin: 40px;
```

```
    background-color: #f4f4f9;
```

```
    color: #333;
```

```
}
```

```
.container {
```

```
    max-width: 600px;
```

```
    background: #fff;
```

```
    padding: 25px;
```



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```
border-radius: 8px;

box-shadow: 0 2px 5px rgba(0,0,0,0.1);

}

h1 {

  color: #1a365d;

}

.btn {

  display: inline-block;

  margin-top: 15px;

  padding: 10px 15px;

  background-color: #3182ce;

  color: #fff;

  text-decoration: none;

  border-radius: 5px;

}

.btn:hover {

  background-color: #2b6cb0;

}

</style>

</head>

<body>
```



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```
<div class="container">
```

```
<h1>Department of Computer Science & Engineering</h1>
```

```
<p>Welcome to the official department portal of SIT Nagpur. We offer state-of-the-art facilities and top-tier academic guidance to shape the innovators of tomorrow.</p>
```

```
<!-- Linking the student information page -->
```

```
<a href="CS1_Stud.html" class="btn">View Student Information</a>
```

```
</div>
```

```
<!-- JavaScript Program to display the Alert Box -->
```

```
<script>
```

```
    window.onload = function() {
```

```
        alert("Welcome to SIT Nagpur");
```

```
    };
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



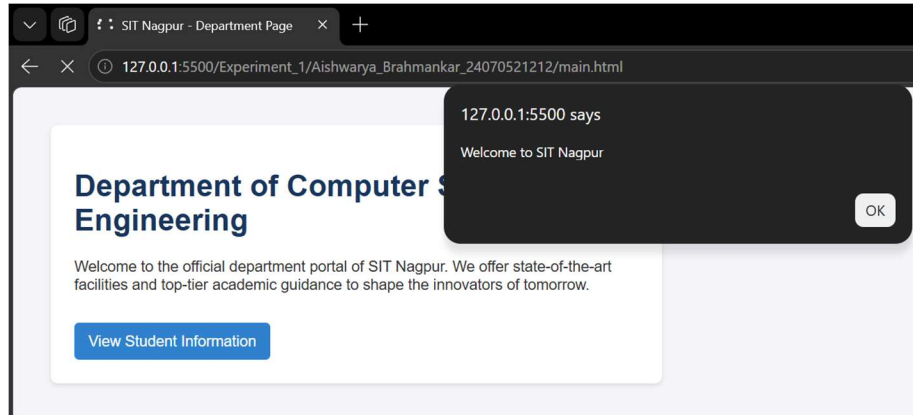
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2. Student:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Student Information</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
margin: 40px;
```

```
background-color: #f4f4f9;
```

```
color: #333;
```

```
}
```

```
.container {
```

```
max-width: 600px;
```

```
background: #fff;
```



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```
padding: 25px;

border-radius: 8px;

box-shadow: 0 2px 5px rgba(0,0,0,0.1);

}

h2 {

    color: #1a365d;

}

table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;

}

th, td {

    padding: 10px;

    border: 1px solid #ddd;

    text-align: left;

}

th {

    background-color: #edf2f7;

}

.back-link {

    display: inline-block;
```



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```
margin-top: 20px;

color: #3182ce;

text-decoration: none;

}

.back-link:hover {

    text-decoration: underline;

}

</style>

</head>

<body>

<div class="container">

    <h2>Student Information Page</h2>

    <table>

        <thead>

            <tr>

                <th>Roll No</th>

                <th>Student Name</th>

                <th>Year</th>

            </tr>

        </thead>
```



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```
<tbody>

<tr>

<td>24070521212</td>

<td>Aishwarya Brahmankar</td>

<td>3rd Year</td>

</tr>

</tbody>

</table>

<!-- Link back to Department page -->

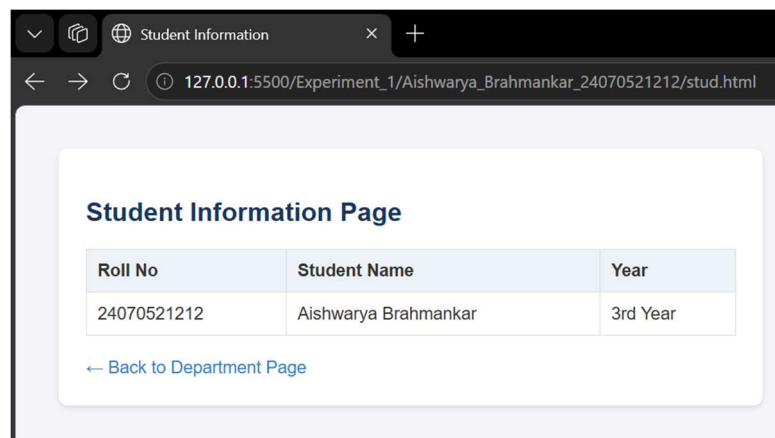
<a href="CS1_Main.html" class="back-link">← Back to Department Page</a>

</div>

</body>

</html>
```

Output:





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Experiment No.: 02

Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Billing Calculator</title>
```

```
</head>
```

```
<body>
```

```
<h2>Billing Calculator</h2>
```

```
<script>
```

```
const GST = 0.18;
```



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```
var customerName = prompt("Enter Customer Name: ");
```

```
var productName = prompt("Enter Product Name: ");
```

```
var quantity = Number(prompt("Enter quantity"));
```

```
let price = Number(prompt("Enter price per Item"));
```

```
let total = quantity * price;
```

```
let gstAmount = total * GST;
```

```
let finalBill = total + gstAmount;
```

```
const bill = {
```

```
    customer: customerName,
```

```
    product: productName,
```

```
    quantity,
```

```
    price,
```

```
    total,
```

```
    gstAmount,
```

```
    finalBill
```

```
};
```



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```
const {  
  
    customer,  
  
    product,  
  
    gstAmount: gst,  
  
    finalBill: amount  
}  
= bill;  
  
document.write(`<h3>Billing Receipt</h3>`);  
  
document.write(`<p>Customer: ${customer}</p>`);  
  
document.write(`<p>Product: ${product}</p>`);  
  
document.write(`<p>Quantity: ${quantity}</p>`);  
  
document.write(`<p>Price per Item: ₹${price}</p>`);  
  
document.write(`<p>Total: ₹${total}</p>`);  
  
document.write(`<p>GST (18%): ₹${gst.toFixed(2)}</p>`);  
  
document.write(`<h3>Final Bill: ₹${amount.toFixed(2)}</h3>`);  
  
</script>
```



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</body>

</html>

Output:

▼

📁

🌐 Billing Calculator

×

+

←

↻

📄 127.0.0.1:5500/practical_2.html

Billing Calculator

Billing Receipt

Customer: Aishwarya Brahmanekar

Product: Dell laptop

Quantity: 1

Price per Item: ₹24070521212

Total: ₹24070521212

GST (18%): ₹4332693818.16

Final Bill: ₹28403215030.16



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Case Study No.: 02

Title: To design and develop an interactive, client-side billing application (BILL.HTML) using HTML5, CSS3, and JavaScript (ES6+) that parses real-time product quantities, applies discount thresholds, calculates taxation rates, and renders an itemised invoice directly onto the Document Object Model (DOM).

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>E-Commerce Automated Billing Engine</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
      line-height: 1.6;
```

```
      margin: 0;
```

```
      padding: 30px;
```

```
      background-color: #f7f9fc;
```



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}

.billing-container {

max-width: 700px;

background: #ffffff;

margin: auto;

padding: 25px;

border-radius: 8px;

box-shadow: 0 4px 15px rgba(0,0,0,0.05);

}

h2, h3 {

color: #1a365d;

text-align: center;

margin-top: 0;

}

.student-badge {

text-align: center;

font-size: 14px;

color: #718096;

margin-bottom: 25px;



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border-bottom: 2px dashed #e2e8f0;

padding-bottom: 10px;

}

.form-row {

display: flex;

gap: 15px;

margin-bottom: 15px;

}

.form-group {

flex: 1;

display: flex;

flex-direction: column;

}

label {

font-weight: 600;

margin-bottom: 6px;

color: #2d3748;

}

input, select {



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```
padding: 10px;

border: 1px solid #cbd5e0;

border-radius: 5px;

font-size: 15px;

}

input:focus, select:focus {

    outline: none;

    border-color: #3182ce;

}

button {

    background: #2b6cb0;

    color: #ffffff;

    padding: 12px;

    border: none;

    border-radius: 5px;

    cursor: pointer;

    width: 100%;

    font-size: 16px;

    font-weight: bold;
```



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```
margin-top: 10px;

transition: background 0.2s;

}

button:hover {

    background: #2c5282;

}

.invoice-box {

    margin-top: 25px;

    padding: 20px;

    background: #f8f9fc;

    border: 1px dashed #4299e1;

    border-radius: 6px;

    display: none;

}

.invoice-table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;

}
```



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```
.invoice-table th, .invoice-table td {
```

```
    padding: 10px;
```

```
    text-align: left;
```

```
    border-bottom: 1px solid #e2e8f0;
```

```
}
```

```
.invoice-table th {
```

```
    background: #edf2f7;
```

```
    color: #2d3748;
```

```
}
```

```
.text-right {
```

```
    text-align: right !important;
```

```
}
```

```
.total-highlight {
```

```
    font-size: 18px;
```

```
    font-weight: bold;
```

```
    color: #e53e3e;
```

```
}
```

```
</style>
```

```
</head>
```



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<body>

<div class="billing-container">

<h2>E-Commerce Billing System</h2>

<div class="student-badge">Developer: Aishwarya Brahmanekar | PRN: 24070521212</div>

<form id="billingForm" onsubmit="generateInvoice(event)">

<div class="form-row">

<div class="form-group">

<label for="customerName">Customer Name:</label>

<input type="text" id="customerName" placeholder="Enter name" required>

</div>

<div class="form-group">

<label for="invoiceNo">Invoice Number:</label>

<input type="text" id="invoiceNo" placeholder="INV-2026-001" required>

</div>

</div>

<div class="form-row">



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```
<div class="form-group" style="flex: 2;">
```

```
<label for="itemName">Item Description:</label>
```

```
<input type="text" id="itemName" placeholder="Product name/model" required>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="itemCategory">Category:</label>
```

```
<select id="itemCategory">
```

```
<option value="Electronics">Electronics (18% GST)</option>
```

```
<option value="Clothing">Clothing (5% GST)</option>
```

```
<option value="Essentials">Essentials (0% GST)</option>
```

```
</select>
```

```
</div>
```

```
</div>
```

```
<div class="form-row">
```

```
<div class="form-group">
```

```
<label for="unitPrice">Unit Price (₹):</label>
```

```
<input type="number" id="unitPrice" step="0.01" min="0" placeholder="0.00"
required>
```



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```
</div>

<div class="form-group">

  <label for="quantity">Quantity:</label>

  <input type="number" id="quantity" min="1" placeholder="1" required>

</div>

<div class="form-group">

  <label for="discountRate">Discount (%):</label>

  <input type="number" id="discountRate" min="0" max="100" placeholder="0">

</div>

</div>

<button type="submit">Compute Calculations & Generate Receipt</button>

</form>

<div id="invoiceBox" class="invoice-box">

  <h3>RETAIL INVOICE RECEIPT</h3>

  <p><strong>Customer Reference:</strong> <span id="invCustomer"></span><br>

  <strong>Invoice String ID:</strong> <span id="invId"></span></p>
```



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<table class="invoice-table">

<thead>

<tr>

<th>Item Description</th>

<th class="text-right">Rate</th>

<th class="text-right">Qty</th>

<th class="text-right">Subtotal</th>

</tr>

</thead>

<tbody>

<tr>

<td> ()</td>

<td class="text-right">₹</td>

<td class="text-right"></td>

<td class="text-right">₹</td>

</tr>

<tr>

<td colspan="3" class="text-right">Discount Applied:</td>

<td class="text-right" style="color: #38a169;">-₹</td>



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</tr>

<tr>

<td colspan="3" class="text-right">Calculated Tax Component (GST):</td>

<td class="text-right">₹</td>

</tr>

<tr style="background: #f7fafc;">

<td colspan="3" class="text-right total-highlight">Grand Total

Payable:</td>

<td class="text-right total-highlight">₹</td>

</tr>

</tbody>

</table>

</div>

</div>

<script>

function generateInvoice(event) {

event.preventDefault();



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// 1. Data Capture Extractors

```
var customer = document.getElementById("customerName").value.trim();

var invoiceNum = document.getElementById("invoiceNo").value.trim();

var item = document.getElementById("itemName").value.trim();

var category = document.getElementById("itemCategory").value;

var price = parseFloat(document.getElementById("unitPrice").value);

var qty = parseInt(document.getElementById("quantity").value);

var discRate = parseFloat(document.getElementById("discountRate").value) || 0;
```

// 2. Arithmetic Matrix Calculations

```
var rawSubtotal = price * qty;

var discountAmount = (discRate / 100) * rawSubtotal;

var discountedSubtotal = rawSubtotal - discountAmount;
```

// Tax Percentage Matrix Assignment

```
var taxPercentage = 0;

if (category === "Electronics") taxPercentage = 18;

else if (category === "Clothing") taxPercentage = 5;

else if (category === "Essentials") taxPercentage = 0;
```



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```
var taxAmount = (taxPercentage / 100) * discountedSubtotal;
```

```
var grandTotal = discountedSubtotal + taxAmount;
```

```
// 3. UI Node Rendering Updates
```

```
document.getElementById("invCustomer").innerText = customer;
```

```
document.getElementById("invId").innerText = invoiceNum;
```

```
document.getElementById("invItem").innerText = item;
```

```
document.getElementById("invCat").innerText = category;
```

```
document.getElementById("invPrice").innerText = price.toFixed(2);
```

```
document.getElementById("invQty").innerText = qty;
```

```
document.getElementById("invSub").innerText = rawSubtotal.toFixed(2);
```

```
document.getElementById("invDisc").innerText = discountAmount.toFixed(2);
```

```
document.getElementById("invTax").innerText = taxAmount.toFixed(2);
```

```
document.getElementById("invGrand").innerText = grandTotal.toFixed(2);
```

```
// Slide/Toggle element block presence configurations
```

```
document.getElementById("invoiceBox").style.display = "block";
```

```
}
```




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Experiment No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>

<head>

<title>JS Practical - Multi-Subject Grading System</title>

<style>

body {

    font-family: Arial, sans-serif;

    max-width: 480px;

    margin: 30px auto;

    padding: 20px;

    border: 1px solid #ccc;

    border-radius: 8px;

    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);

}

.form-group {

    margin-bottom: 15px;

}
```



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```
label {  
  
    display: block;  
  
    margin-bottom: 4px;  
  
    font-weight: bold;  
  
}  
  
input {  
  
    width: 100%;  
  
    padding: 8px;  
  
    box-sizing: border-box;  
  
    border: 1px solid #ccc;  
  
    border-radius: 4px;  
  
    outline: none;  
  
}  
  
/* Highlight invalid input fields */  
  
input.invalid-input {  
  
    border: 2px solid #dc3545;  
  
    background-color: #fff8f8;  
  
}  
  
/* Style inline error text under inputs */  
  
.field-error {  
  
    color: #dc3545;  
  
    font-size: 13px;
```



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```
margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */

}

button {

  width: 100%;

  padding: 10px;

  background-color: #007bff;

  color: white;

  border: none;

  border-radius: 4px;

  cursor: pointer;

  font-size: 16px;

  margin-top: 10px;

}

button:hover {

  background-color: #0056b3;

}

#result {

  margin-top: 20px;

  padding: 12px;

  border-radius: 4px;
```



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```
display: none;

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}

</style>

</head>

<body>

<h2>Grading System</h2>

<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>

  <div class="form-group">

    <label for="studentName">Student Name:</label>

    <input type="text" id="studentName" placeholder="e.g. John Doe">

    <span class="field-error" id="nameError"></span>

  </div>

  <div class="form-group">

    <label for="math">Math Marks (0 - 100):</label>

    <input type="number" id="math" placeholder="e.g. 85">
```



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</div>

<div class="form-group">

<label for="science">Science Marks (0 - 100):</label>

<input type="number" id="science" placeholder="e.g. 90">

</div>

<div class="form-group">

<label for="english">English Marks (0 - 100):</label>

<input type="number" id="english" placeholder="e.g. 78">

</div>

<button type="submit">Calculate Overall Grade</button>

</form>

<div id="result"></div>

<script>

function calculateAverageGrade(event) {



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```
event.preventDefault();
```

```
// Track whether the form is completely valid
```

```
let isValid = true;
```

```
// Reset previous result box
```

```
const resultDiv = document.getElementById("result");
```

```
resultDiv.style.display = "none";
```

```
// -----
```

```
// Helper Function: Validate an individual field
```

```
// -----
```

```
function validateField(inputId, errorId, fieldName, isSubject = true) {
```

```
    const inputElement = document.getElementById(inputId);
```

```
    const errorElement = document.getElementById(errorId);
```

```
    const value = inputElement.value.trim();
```

```
// Clear previous styles/messages
```

```
inputElement.classList.remove("invalid-input");
```

```
errorElement.textContent = "";
```

```
// Rule 1: Empty text validation
```



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```
if (value === "") {  
    inputElement.classList.add("invalid-input");  
    errorElement.textContent = `Please enter ${fieldName}.`;   
    isValid = false;  
    return;  
}  
  
// Rule 2: Numeric & Range validation (for subject marks)  
if (isSubject) {  
    const score = parseFloat(value);  
    if (isNaN(score) || score < 0 || score > 100) {  
        inputElement.classList.add("invalid-input");  
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;   
        isValid = false;  
    }  
}  
  
// -----  
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY  
// -----  
  
validateField("studentName", "nameError", "student's name", false);
```



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```
validateField("math", "mathError", "Math marks", true);

validateField("science", "scienceError", "Science marks", true);

validateField("english", "englishError", "English marks", true);


// Stop calculation if any field failed validation
if (!isValid) return;


// -----

// 2. CALCULATE GRADE (Runs only if all fields are valid)

// -----

const math = parseFloat(document.getElementById("math").value);
const science = parseFloat(document.getElementById("science").value);
const english = parseFloat(document.getElementById("english").value);
const name = document.getElementById("studentName").value.trim();


const totalMarks = math + science + english;
const average = (totalMarks / 3).toFixed(2);


let grade = "";
let status = "";


if (average >= 90) {
```



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```
grade = "A+"; status = "Passed with Distinction";

} else if (average >= 80) {

    grade = "A"; status = "Passed";

} else if (average >= 70) {

    grade = "B"; status = "Passed";

} else if (average >= 60) {

    grade = "C"; status = "Passed";

} else if (average >= 50) {

    grade = "D"; status = "Passed";

} else {

    grade = "F"; status = "Failed";

}

// Display result summary

resultDiv.style.display = "block";

resultDiv.innerHTML = `

    <strong>Result Summary:</strong><br>

    Name: ${name}<br>

    Math: ${math} | Science: ${science} | English: ${english}<br>

    Total: ${totalMarks} / 300<br>

    Average: <strong>${average}%</strong><br>

    Overall Grade: <strong>${grade}</strong> (${status})
```



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```
`;  
}  
  
</script>
```

```
</body>
```

```
</html>
```

Output:

Grading System

Student Name:
Aishwarya Brahmkar

Math Marks (0 - 100):
95

Science Marks (0 - 100):
89

English Marks (0 - 100):
97

Calculate Overall Grade

Result Summary:
Name: Aishwarya Brahmkar
Math: 95 | Science: 89 | English: 97
Total: 281 / 300
Average: 93.67%
Overall Grade: A+ (Passed with Distinction)



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Case Study No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>
```

```
<head>
```

```
<title>JS Practical - Multi-Subject Grading System</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    max-width: 480px;
```

```
    margin: 30px auto;
```

```
    padding: 20px;
```

```
    border: 1px solid #ccc;
```

```
    border-radius: 8px;
```

```
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
```

```
}
```

```
.form-group {
```



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margin-bottom: 15px;

}

label {

display: block;

margin-bottom: 4px;

font-weight: bold;

}

input {

width: 100%;

padding: 8px;

box-sizing: border-box;

border: 1px solid #ccc;

border-radius: 4px;

outline: none;

}

/* Highlight invalid input fields */

input.invalid-input {

border: 2px solid #dc3545;

background-color: #fff8f8;



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}

/* Style inline error text under inputs */

.field-error {

color: #dc3545;

font-size: 13px;

margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */

}

button {

width: 100%;

padding: 10px;

background-color: #007bff;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 16px;

margin-top: 10px;



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```
}
```

```
button:hover {
```

```
background-color: #0056b3;
```

```
}
```

```
#result {
```

```
margin-top: 20px;
```

```
padding: 12px;
```

```
border-radius: 4px;
```

```
display: none;
```

```
background-color: #d4edda;
```

```
color: #155724;
```

```
border: 1px solid #c3e6cb;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Grading System</h2>
```



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```
<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>
```

```
<div class="form-group">
```

```
<label for="studentName">Student Name:</label>
```

```
<input type="text" id="studentName" placeholder="e.g. John Doe">
```

```
<span class="field-error" id="nameError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="math">Math Marks (0 - 100):</label>
```

```
<input type="number" id="math" placeholder="e.g. 85">
```

```
<span class="field-error" id="mathError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="science">Science Marks (0 - 100):</label>
```

```
<input type="number" id="science" placeholder="e.g. 90">
```

```
<span class="field-error" id="scienceError"></span>
```

```
</div>
```



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```
<div class="form-group">
```

```
<label for="english">English Marks (0 - 100):</label>
```

```
<input type="number" id="english" placeholder="e.g. 78">
```

```
<span class="field-error" id="englishError"></span>
```

```
</div>
```

```
<button type="submit">Calculate Overall Grade</button>
```

```
</form>
```

```
<div id="result"></div>
```

```
<script>
```

```
function calculateAverageGrade(event) {
```

```
    event.preventDefault();
```

```
    // Track whether the form is completely valid
```

```
    let isValid = true;
```



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```
// Reset previous result box

const resultDiv = document.getElementById("result");

resultDiv.style.display = "none";


// -----

// Helper Function: Validate an individual field

// -----

function validateField(inputId, errorId, fieldName, isSubject = true) {

    const inputElement = document.getElementById(inputId);

    const errorElement = document.getElementById(errorId);

    const value = inputElement.value.trim();

    // Clear previous styles/messages

    inputElement.classList.remove("invalid-input");

    errorElement.textContent = "";

    // Rule 1: Empty text validation

    if (value === "") {

        inputElement.classList.add("invalid-input");
```



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```
errorElement.textContent = `Please enter ${fieldName}.`;
```

```
isValid = false;
```

```
return;
```

```
}
```

```
// Rule 2: Numeric & Range validation (for subject marks)
```

```
if (isSubject) {
```

```
    const score = parseFloat(value);
```

```
    if (isNaN(score) || score < 0 || score > 100) {
```

```
        inputElement.classList.add("invalid-input");
```

```
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;
```

```
        isValid = false;
```

```
    }
```

```
}
```

```
}
```

```
// -----
```

```
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY
```

```
// -----
```



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```
validateField("studentName", "nameError", "student's name", false);
```

```
validateField("math", "mathError", "Math marks", true);
```

```
validateField("science", "scienceError", "Science marks", true);
```

```
validateField("english", "englishError", "English marks", true);
```

```
// Stop calculation if any field failed validation
```

```
if (!isValid) return;
```

```
// -----
```

```
// 2. CALCULATE GRADE (Runs only if all fields are valid)
```

```
// -----
```

```
const math = parseFloat(document.getElementById("math").value);
```

```
const science = parseFloat(document.getElementById("science").value);
```

```
const english = parseFloat(document.getElementById("english").value);
```

```
const name = document.getElementById("studentName").value.trim();
```

```
const totalMarks = math + science + english;
```

```
const average = (totalMarks / 3).toFixed(2);
```



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```
let grade = "";

let status = "";

let scholarship = "";

if (average >= 95) {

    grade = "A+"; status = "Passed with Distinction"; scholarship = "Eligible for Platinum
Scholarship";

} else if (average >= 85) {

    grade = "A"; status = "Passed"; scholarship = "Eligible for Gold Scholarship";

} else if (average >= 70) {

    grade = "B"; status = "Passed"; scholarship = "Eligible for Silver Scholarship";

} else if (average >= 50) {

    grade = "C"; status = "Passed"; scholarship = "Eligible for Bronze Scholarship";

} else if (average < 50) {

    grade = "D"; status = "Passed"; scholarship = "No Scholarship";

} else {

    grade = "F"; status = "Failed";

}
```



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// Display result summary

```
resultDiv.style.display = "block";
```

```
resultDiv.innerHTML = `
```

```
    <strong>Result Summary:</strong><br>
```

```
    Name: ${name}<br>
```

```
    Math: ${math} | Science: ${science} | English: ${english}<br>
```

```
    Total: ${totalMarks} / 300<br>
```

```
    Average: <strong>${average}%</strong><br>
```

```
    Overall Grade: <strong>${grade}</strong> (${status}) ${scholarship}
```

```
`;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



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JS Practical - Multi-Subject Grading System

127.0.0.1:5500/Aishwarya_Brahmankar_24070521212/Case%20Study%203/scholarship_grading_system.html

Grading System

Student Name:

Math Marks (0 - 100):

Science Marks (0 - 100):

English Marks (0 - 100):

[Calculate Overall Grade](#)

Result Summary:

Name: Aishwarya Brahmankar

Math: 95 | Science: 89 | English: 97

Total: 281 / 300

Average: 93.67%

Overall Grade: A (Passed) Eligible for Gold Scholarship



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Experiment No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
let welcome = function() {  
    console.log("Welcome to the Palindrome Checker!");  
};  
  
welcome();  
  
// Global variable  
let language = "JavaScript";  
  
// Main function to check if a string or number is a palindrome  
function isPalindrome(str) {  
    // 1. Accept numbers by converting them to strings  
    if (typeof str === "number") {  
        str = String(str);  
    }  
  
    // 2. Validate input type (reject null, undefined, objects, arrays, etc.)
```



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```
if (typeof str !== "string") {  
    throw new TypeError(`Expected a string or number, but received ${typeof str}`);  
}  
  
// 3. Check for empty string  
if (str.trim().length === 0) {  
    throw new Error("Input string cannot be empty.");  
}  
  
// Clean the string: remove non-alphanumeric characters and convert to lowercase  
let cleaned = str.toLowerCase().replace(/[^a-z0-9]/g, "");  
  
// Handle edge case where input had only punctuation/spaces (e.g., "!!!")  
if (cleaned.length === 0) {  
    throw new Error("Input must contain at least one alphanumeric character.");  
}  
  
// Reverse and check  
let reversed = cleaned.split("").reverse().join("");  
return cleaned === reversed;  
}
```



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// Helper function with try...catch error handling

```
function testPalindrome(phrase) {
```

```
  try {
```

```
    let result = isPalindrome(phrase);
```

```
    console.log(` SUCCESS: "${phrase}" -> ${result ? "Is a palindrome!" : "Not a  
palindrome."}`);
```

```
  } catch (error) {
```

```
    // Handle caught errors safely without stopping execution
```

```
    console.error(` ERROR [${error.name}]: Failed to check input (${phrase}) ->  
${error.message}`);
```

```
  }
```

```
}
```

// --- Testing Examples ---

```
console.log(`\nRunning checks using ${language}:\n` + "=".repeat(45));
```

// Valid Test Cases (Strings & Numbers)

```
testPalindrome("Hello World");
```

```
testPalindrome("Aishwarya Brahmkar");
```

```
testPalindrome("24070521212");
```

```
testPalindrome(12321);           // NOW WORKS: Passed as a Number directly!
```

```
testPalindrome(21212);           // NOW WORKS: Evaluates as "Not a palindrome"
```



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// Error Handling Test Cases (Triggers Catch Block)

testPalindrome(""); // Error: Empty string

testPalindrome(null); // TypeError: null passed

testPalindrome(" "); // Error: Whitespace only

testPalindrome("!!!?"); // Error: No alphanumeric characters

Output:

```
[Running] node "e:\JS lab\Experiment_4\palindrome_checker.js"
Welcome to the Palindrome Checker!
ERROR [Error]: Failed to check input () -> Input string cannot be empty.
ERROR [TypeError]: Failed to check input (null) -> Expected a string or number, but received object
ERROR [Error]: Failed to check input ( ) -> Input string cannot be empty.

Running checks using JavaScript:
=====
SUCCESS: "Hello World" -> Not a palindrome.
SUCCESS: "Aishwarya Brahmkar" -> Not a palindrome.
SUCCESS: "24070521212" -> Not a palindrome.
SUCCESS: "12321" -> Is a palindrome!
SUCCESS: "21212" -> Is a palindrome!
ERROR [Error]: Failed to check input (!!!?) -> Input must contain at least one alphanumeric character.

[Done] exited with code=0 in 0.158 seconds
```



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Case Study No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
const readline = require('readline');
```

```
// 1. FUNCTION DECLARATION (Global Scope)
```

```
// Utility function to reverse the entered PIN
```

```
function reversePin(pin) {  
    const pinStr = String(pin);  
    return pinStr.split("").reverse().join("");  
}
```

```
// Helper function to ask a question in the terminal using built-in Node readline
```

```
const askQuestion = (query) => {  
    const rl = readline.createInterface({  
        input: process.stdin,  
        output: process.stdout  
    });  
    return new Promise(resolve => rl.question(query, answer => {
```



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```
rl.close();

resolve(answer);

});

};

// 2. CLOSURE & SCOPE DEMONSTRATION

// ATM Session Generator returning an Arrow Function

const createATMSession = (accountHolder) => {

  // Outer Lexical Scope variable

  const user = accountHolder;

  let attempts = 0;

  // 3. ARROW FUNCTION (Closure over `user` and `attempts`)

  const verifyPinInteractive = async () => {

    attempts++;

    // Dynamic prompt to ask the user for PIN input

    const enteredPin = await askQuestion(`[ATM System] ${user}, please enter any PIN: `);
```



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```
if (!enteredPin || enteredPin.trim() === "") {  
    return `Transaction cancelled by ${user}.`;  
}  
  
const strEnteredPin = String(enteredPin.trim());  
const reversed = reversePin(strEnteredPin); // Calls global Function Declaration  
  
console.log(`\n--- Transaction Attempt #${attempts} for ${user} ---`);  
console.log(`Entered PIN : ${strEnteredPin}`);  
console.log(`Reversed PIN: ${reversed}`);  
  
// Check if the entered PIN is a Palindrome  
const isPalindrome = (strEnteredPin === reversed);  
  
// Access is ALWAYS granted, but security message depends on Palindrome check  
if (isPalindrome) {  
    return `ACCESS GRANTED!\n SPECIAL SECURITY ALERT: PIN "${strEnteredPin}"  
is a Palindrome! Please consider changing your password for better security.`;  
} else {  
    return `ACCESS GRANTED! Welcome, ${user}.`;  
}  
};
```



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```
    return verifyPinInteractive;
};

// 4. ATM SYSTEM EXECUTION

(async () => {

    const johnSession = createATMSession("John Doe");

    // Prompts user in the integrated terminal

    const result = await johnSession();

    console.log("\n" + result);

})();

);
```

Output:

```
PS E:\JS lab> node "e:\JS lab\Aishwarya_Brahmankar_24070521212\Case Study 4\ATM_pin_verification.js"
[ATM System] John Doe, please enter any PIN: 1221

--- Transaction Attempt #1 for John Doe ---
Entered PIN : 1221
Reversed PIN: 1221

ACCESS GRANTED!
SPECIAL SECURITY ALERT: PIN "1221" is a Palindrome! Please consider changing your password for better security.
```



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